

UNIT-III

USE OF NATURAL RESOURCES AND DISASTER MANAGEMENT

Natural resources form the foundation of human civilization. They supply the raw materials for our industries, the food and water we consume, and the energy that propels us forward. However, irresponsible extraction and consumption of these resources has resulted in environmental degradation, biodiversity loss, and climate change, all of which make communities more vulnerable to natural disasters. Floods, droughts, hurricanes, and wildfires are becoming increasingly often and severe, frequently with disastrous effects for people who are unprepared. This unit discusses diverse natural resources, their use, and disaster management.

3.1 Water: Use and over-exploitation of surface and ground water

Water is an extremely valuable renewable resource. Water sustains biodiversity. It has an impact on the local climate. Oceans cover 71% of the earth's surface and account for 97.4% of its water. Overuse of water in irrigation can lead to soil erosion and degradation. Thus proper water management is important.

Uses of Surface and Groundwater:

Surface and groundwater are used for a wide range of purposes:

1. Domestic use
2. Agriculture
3. Industrial use
4. Energy production
5. Recreation and tourism
6. Ecosystem Services

Overexploitation of Surface and Groundwater:

The overexploitation of surface and groundwater resources has serious environmental, economic, and social implications.

1. Depletion of Water Sources:

- Surface Water: Excessive withdrawal from rivers and lakes can deplete water levels, harming ecosystems and human societies. Examples include rivers drying up and lakes receding, such as the Aral Sea.
- Ground Water: Over-pumping of aquifers causes diminishing water tables, making groundwater access more difficult and costly. Regions such as North India and California's Central Valley are experiencing serious groundwater depletion.

2. Water Quality Degradation:

- Surface Water: Pollution from agricultural runoff, industrial discharge, and untreated sewage contaminates surface water bodies, making them hazardous for human use and harmful to aquatic life.

- **Ground Water:** Pesticides, nitrates, heavy metals, and other contaminants can contaminate groundwater, making it unfit for human consumption. Once contaminated, groundwater is tough to clean up.

3. **Reduced Agricultural Productivity:**

Overexploitation of water resources can cause water shortages, limiting irrigation and lowering crop production. Food security and livelihoods are put at risk, particularly in countries that rely heavily on agriculture.

4. **Ecological Impacts:**

Aquatic ecosystems are impacted by lower water levels in rivers and lakes, resulting in the loss of biodiversity and interruption of natural processes. The decline of wetlands diminishes the ability of these ecosystems to offer crucial services like flood control and water filtration.

5. **Economic cost:**

The economic costs of water overexploitation include increased expenditure on water extraction, treatment, and infrastructure. Industries and municipalities may face higher costs for accessing and purifying water.

6. **Social Consequences:**

Water scarcity and contamination can cause conflicts over water supplies, community dislocation, and negative health outcomes due to a lack of access to safe drinking water.

Conflicts over Water: International and Inter-State Conflict:

Water conflicts arise when the demand for water exceeds supply, and competing interests lead to disputes. These conflicts can occur at various scales, including international and inter-state levels. Understanding the root causes and dynamics of these conflicts is crucial for developing effective management and resolution strategies.

Causes of water Conflicts:

Water conflicts are driven by several factors, including:

- **Scarcity:** Limited water resources coupled with high demand can lead to competition and conflict.
- **Geopolitical Boundaries:** Rivers and lakes often cross political borders, creating disputes over water rights and usage.
- **Population Growth:** Increasing populations heighten demand for water, exacerbating existing tensions.
- **Economic Development:** Industrialization and agricultural expansion increase water consumption, potentially at the expense of neighbouring regions or countries.
- **Climate Change:** Altered precipitation patterns, more frequent droughts, and changing river flows due to climate change can intensify water disputes.
- **Pollution:** Contamination of water sources can reduce the availability of clean water, leading to conflicts over remaining uncontaminated supplies.

International Water Conflicts:

International water conflicts occur when countries share trans boundary water bodies. These conflicts frequently include complex geopolitical, legal, and environmental considerations. Following are the examples of international water conflicts:

- Dispute over water in the Nile Basin
- China-India: The Brahmaputra River

Inter-State Water Conflicts:

Inter-state water conflicts arise within a single country, usually between states or regions that share water resources. These disagreements can be as serious as international ones, particularly in countries with federal systems of government. Following are the examples of international water conflicts:

- Cauvery water dispute: Between Tamil Nadu and Karnataka this is a crucial issue
- Narmada water dispute: It is among Gujarat, Madhya Pradesh, and Maharashtra.

Land Resources and Land Use Change:

Land is crucial for human survival and progress, serving as the foundation for farming, housing, industrial activities, and natural ecosystems. The management and utilization of land have significant effects on environmental sustainability, economic development, and societal

welfare. This section delves into the different facets of land resources, the factors and consequences of changes in land use, and approaches for responsible land management.

Types of Natural Resources:

- **Agricultural Land:** Areas used for growing crops and raising livestock.
- **Forests:** Land covered with trees and other vegetation, providing timber, habitat, and ecosystem services.
- **Urban Land:** Areas developed for human habitation, including cities and towns.
- **Wetlands:** Land areas saturated with water, such as swamps and marshes, which are critical for biodiversity and water purification.

Uses of Land Resources:

- **Agriculture:** Growing food, fiber, and biofuels.
- **Forestry:** Harvesting timber and non-timber forest products.
- **Industry and Infrastructure:** Developing factories, roads, and buildings.
- **Recreation and Conservation:** Preserving natural areas for leisure and biodiversity.

Land Use Changes

Land use change refers to the transformation of the natural landscape, often due to human activities, which alters the land's natural state and function.

Causes of Land Use Change:

- **Urbanization:** Expansion of cities and towns into rural areas, often leading to the conversion of agricultural and natural lands.
- **Agricultural Expansion:** Conversion of forests and grasslands into farmland to meet the growing demand for food and biofuels.
- **Deforestation:** Clearing forests for timber, agriculture, and development.
- **Infrastructure Development:** Building roads, highways, and other infrastructure that disrupts natural landscapes.

Impacts of Land Use Change:

- **Environmental Degradation:** Loss of biodiversity, soil erosion, and disruption of water cycles.
- **Climate Change:** Release of carbon dioxide from deforestation and soil disturbance, contributing to global warming.
- **Habitat Loss:** Destruction of natural habitats, leading to species extinction.
- **Water Quality and Availability:** Altered water runoff patterns and pollution from agricultural runoff and urban areas.

Land Degradation, Soil Erosion, and Their Effects:

Land Degradation

Land degradation refers to the decline in the quality and productivity of land due to human activities or natural processes. It involves the deterioration of the physical, chemical, and biological properties of the soil and the loss of vegetation cover.

Causes of Land Degradation:

1. **Deforestation:** Clearing forests for agriculture or urban development reduces vegetation cover and disrupts ecosystems.
2. **Overgrazing:** Excessive grazing by livestock leads to the removal of vegetation, soil compaction, and erosion.
3. **Agricultural Practices:** Unsustainable farming methods, such as monocropping, overuse of chemical fertilizers and pesticides, and improper irrigation, degrade soil health.
4. **Urbanization and Infrastructure Development:** Construction activities disturb the soil and vegetation, leading to erosion and loss of fertile land.
5. **Mining:** Extraction of minerals disrupts the soil structure, contaminates water sources, and removes vegetation.
6. **Climate Change:** Extreme weather events, such as droughts and floods, exacerbate land degradation.

Effects of Land Degradation:

1. **Loss of Soil Fertility:** Reduced organic matter and nutrients in the soil lead to lower agricultural productivity.
2. **Biodiversity Loss:** Degraded land cannot support diverse plant and animal species, leading to loss of habitats and species extinction.
3. **Water Scarcity:** Reduced vegetation cover and soil compaction decrease water infiltration and storage, affecting water availability.
4. **Desertification:** Severe land degradation in arid and semi-arid areas can lead to the expansion of deserts.
5. **Economic Impact:** Reduced agricultural yields and increased costs of land restoration affect the livelihoods of farmers and local communities.
6. **Climate Change:** Degraded lands release stored carbon, contributing to increased atmospheric CO₂ levels.

Soil Erosion

Soil erosion is the process by which the topsoil is removed by wind, water, or human activities. It is a major contributor to land degradation and has significant environmental and economic impacts.

Natural Causes of Soil Erosion

1. **Water Erosion:**
 - **Rainfall and Surface Runoff:** Intense and heavy rainfall can dislodge soil particles, leading to sheet, rill, and gully erosion. Surface runoff, the flow of water over the land, carries these particles away.
 - **River and Stream Flow:** The natural flow of rivers and streams can erode their banks and beds, transporting soil downstream.
2. **Wind Erosion:**
 - **High Wind Velocities:** Strong winds can lift and transport soil particles, especially in dry and arid regions where vegetation cover is sparse.
 - **Dry Soil Conditions:** Soil that is dry and loose is more susceptible to wind erosion.
3. **Gravity Erosion:**
 - **Mass Wasting:** Processes such as landslides, rockfalls, and soil creep, where gravity causes the downslope movement of soil and rock.

Human-Induced Causes of Soil Erosion

1. **Agricultural Practices:**
 - **Tillage:** Frequent ploughing and tillage disturb the soil structure, making it more prone to erosion.
 - **Monocropping:** Growing the same crop repeatedly on the same land depletes soil nutrients and reduces organic matter, leading to increased erosion.
 - **Overgrazing:** Excessive grazing by livestock removes vegetation cover and compacts the soil, increasing susceptibility to erosion.
 - **Improper Irrigation:** Over-irrigation can lead to waterlogging and subsequent runoff, while under-irrigation can lead to soil drying and wind erosion.
2. **Deforestation and Vegetation Removal:**
 - **Logging and Land Clearing:** Removing trees and vegetation for timber, agriculture, or development leaves soil exposed to erosion by wind and water.
 - **Slash-and-Burn Agriculture:** Burning vegetation to clear land for farming depletes soil nutrients and leaves it vulnerable to erosion.
3. **Urbanization and Infrastructure Development:**
 - **Construction Activities:** Building roads, houses, and other infrastructure disturbs the soil and removes vegetation, leading to increased erosion.
 - **Impervious Surfaces:** Urban areas with extensive concrete and asphalt surfaces prevent water infiltration, increasing surface runoff and soil erosion.
4. **Mining Activities:**
 - **Surface Mining:** Removing topsoil and vegetation to access minerals disrupts the soil structure and exposes it to erosion.
 - **Open-Pit Mining:** Creates large pits and waste heaps that are prone to erosion.

5. **Climate Change:**

- **Extreme Weather Events:** Increased frequency and intensity of storms, heavy rainfall, and droughts due to climate change exacerbate soil erosion.
- **Sea Level Rise:** Coastal erosion is intensified by rising sea levels and increased storm surges.

6. **Poor Land Management Practices:**

- **Inadequate Soil Conservation Measures:** Lack of practices like contour ploughing, terracing, and cover cropping can lead to increased erosion.
- **Neglect of Marginal Lands:** Cultivating lands that are naturally prone to erosion without appropriate management can worsen soil loss.

Effects of Soil Erosion:

1. **Loss of Topsoil:** The nutrient-rich topsoil is essential for plant growth, and its loss reduces soil fertility and agricultural productivity.
2. **Water Pollution:** Eroded soil particles can carry pesticides and fertilizers into water bodies, leading to contamination and eutrophication.
3. **Sedimentation:** Accumulation of eroded soil in rivers, lakes, and reservoirs reduces water storage capacity and affects aquatic habitats.
4. **Infrastructure Damage:** Erosion can undermine roads, bridges, and buildings, leading to costly repairs and maintenance.
5. **Desertification:** Persistent erosion in arid and semi-arid regions can lead to desert-like conditions.

ENVIRONMENTAL IMPACT ASSESSMENT (EIA)

EIA stands for Environmental Impact Assessment. Simply said, EIA is a technique for evaluating the environmental effect of a planned development. When conducting an Environmental Impact Assessment (EIA), the interrelated socioeconomic, cultural, and human-health aspects are evaluated. This is used to predict the environmental impacts of a project in the pre-planning stage itself so that decisions can be taken to reduce the adverse impacts.

Evolution and History of EIA:

EIA originated in the 1970s. In 1969, the United States enacted the first National Environmental Policy Act (NEPA). EIA was first used by industrialized countries, but it was gradually adopted by developing countries such as India. Columbia and the Philippines were among the first developing countries to incorporate environmental impact assessments into their policy. The primary goal of EIA is to maintain the environment and determine the best balance of economic and environmental costs and benefits.

Environmental Impact Assessment (EIA) Process

Environmental Impact Assessment (EIA) Process	
Process	Details in Brief
Screening	At this point, the projects that require a complete or partial evaluation study are determined.
Scoping	- At this level, decisions are made regarding which impacts require assessment. In the process, public participation, specialist knowledge, international agreements, and legal requirements are taken into account. - During this phase, alternative options that prevent or at least lessen the project's negative effects are also investigated. - Research is done on alternative plans or locations that reduce or eliminate impact.
Assessment and Evaluation of Impact Development of alternatives	The planned project's environmental effects are examined, and the alternatives to such projects are highlighted.
EIA Report	In this step, a non-technical overview of the project's effects and an environmental management plan (EMP) are created for the general public.
Decision Making	The project's destiny has been determined.

	Whether the project is to be given approval or not and if it is to be given, under what conditions
Monitoring, Compliance, enforcement and environmental auditing	observing if the mitigation measures and anticipated effects are carried out in accordance with the EMP

Importance of EIA:

- ❑ **Informed Decision-Making:** Provides decision-makers with comprehensive information about the potential environmental impacts.
- ❑ **Public Participation:** Involves the public and stakeholders, ensuring transparency and inclusion.
- ❑ **Sustainable Development:** Ensures that development projects consider environmental sustainability, reducing adverse impacts on ecosystems.
- ❑ **Legal Compliance:** Helps projects comply with environmental laws and regulations, avoiding legal and financial penalties.

Components of an Environmental Impact Statement (EIS):

- ❑ **Project Description:** Detailed description of the proposed project, including its purpose, location, size, and various components.
- ❑ **Baseline Conditions:** Assessment of the current environmental conditions (e.g., air, water, soil, flora, fauna) in the project area.
- ❑ **Impact Analysis:** Detailed analysis of potential environmental impacts, both positive and negative, direct and indirect, short-term and long-term.
- ❑ **Mitigation Measures:** Specific strategies and actions to prevent, reduce, or manage adverse environmental impacts.
- ❑ **Environmental Management Plan (EMP):** Plan for monitoring and managing environmental impacts throughout the project lifecycle.

Challenges in EIA:

- ❑ **Data Limitations:** Lack of reliable baseline data can hinder accurate impact predictions.
- ❑ **Uncertainties:** Uncertainty in predicting environmental impacts and the effectiveness of mitigation measures.
- ❑ **Public Opposition:** Projects can face significant public opposition, particularly if perceived environmental and social impacts are high.
- ❑ **Complexity:** Large and complex projects can involve numerous environmental, social, and economic factors, making the EIA process complicated.

Environmental Impacts of Dam building:

Construction of a dam has a major effect on the environment, forests, wildlife, and communities of tribal people. Depending on the project's size, location, and management techniques, these effects might be either favourable or bad. Here are thorough justifications of these effects:

Hydrological Changes:

- **Modified Water Flow:** Dams control the flow of rivers, which can impact the dynamics of natural floodplains and lessen seasonal floods downstream.
- **Transport of Sediment:** The dam traps sediment that would normally replenish downstream ecosystems, causing erosion and the loss of nutrient-rich soils downstream.

Water Quality:

- **Temperature Changes:** Water released from dams can be colder or warmer than natural river temperatures, disrupting aquatic ecosystems.
- **Oxygen Levels:** Reduced oxygen levels in reservoir water can harm fish and other aquatic life.

Greenhouse Gas Emissions

- **Methane Production:** Decomposing organic matter in reservoirs can produce methane, a potent greenhouse gas.

Forest Impacts

1. Deforestation

- **Land Clearance:** Large areas of forest are often cleared to create reservoirs, leading to habitat loss and fragmentation.
- **Road Construction:** Building infrastructure for dam construction can also lead to deforestation and increased access for illegal logging.

2. Loss of Ecosystem Services

- **Biodiversity:** Forests provide habitat for numerous species, and their loss can reduce biodiversity.
- **Carbon Sequestration:** Forests act as carbon sinks, and their removal contributes to increased atmospheric CO₂ levels.

Biodiversity Impacts

1. Aquatic Ecosystems

- **Fish Migration:** Dams block migration routes for fish species, such as salmon, disrupting their life cycles and reducing populations.
- **Habitat Alteration:** Reservoirs create still water habitats, replacing flowing river ecosystems and affecting species adapted to specific conditions.

2. Terrestrial Ecosystems

- **Habitat Loss:** Flooding large areas for reservoirs can lead to the loss of terrestrial habitats, affecting plant and animal species.

- **Species Displacement:** Species may be forced to relocate, which can lead to competition and changes in local biodiversity.

Tribal Population Impacts

1. Displacement and Resettlement

- **Forced Relocation:** Indigenous and tribal communities living in dam sites are often displaced, losing their homes and traditional lands.
- **Cultural Loss:** Displacement can lead to the loss of cultural heritage, traditional knowledge, and social structures.

2. Livelihoods

- **Loss of Resources:** Dams can submerge agricultural land, forests, and fishing areas that tribal populations depend on for their livelihoods.
- **Economic Impact:** While some dams provide benefits like electricity and irrigation, displaced communities often face economic hardship due to loss of traditional livelihoods.

3. Health and Social Impacts

- **Psychosocial Stress:** Forced displacement can cause significant stress and mental health issues among tribal populations.
- **Access to Services:** Resettled communities may have limited access to essential services like healthcare, education, and clean water.

Mitigation Measures

- **Fish Passages:** Building canals or ladders to facilitate fish migration. Using procedures for sediment flushing in order to preserve sediment transport downstream is known as sediment management.

- **Preservation of Forests**

Planting trees to make up for deforested regions is known as reforestation programs. The creation of protected areas is necessary to preserve the biodiversity and surviving forests.

☐ **Biodiversity Protection**

- **Habitat Creation:** Creating new habitats to replace those lost to reservoirs.
- **Monitoring Programs:** Continuous monitoring of biodiversity to assess and mitigate impacts.
- **Support for Tribal Populations**
- **Fair Compensation:** Ensuring fair and adequate compensation for displaced communities.

- **Community Involvement:** Involving tribal communities in planning and decision-making processes.
- **Livelihood Restoration:** Providing alternative livelihood options and training programs for displaced populations.

Disaster Management

Disaster management involves a systematic approach to preparing for, responding to, and recovering from disasters. This topics involves various types of disasters like floods,droughts,earthquakes,cyclones and landslides.

What is Disaster?

A disaster is defined as a disruption on a massive scale, either natural or man-made, occurring in short or long periods. Disasters can lead to human, material, economic or environmental hardships, which can be beyond the bearable capacity of the affected society. As per statistics, India as a whole is vulnerable to 30 different types of disasters that will affect the economic, social, and human development potential to such an extent that it will have long-term effects on productivity and macro-economic performance.

Flood and Drought Management:

Floods and droughts are two extreme weather events that require distinct management strategies due to their differing natures and impacts. Below is an overview of management practices for each

Floods:

Floods are natural disasters that occur when an overflow of water submerges land that is usually dry. Flooding can happen in various ways and can be triggered by a number of factors. Here are the primary causes of floods:

1. Heavy Rainfall
2. Snow melt
3. River overflow
4. Coastal flooding(Storm surges and Tsunami)
5. Dam failure
6. Urban flooding due to inadequate drainage

Effects of flood:

- ☐ **Human Impact:** Loss of life, injuries, displacement of people, and health issues due to waterborne diseases.
- ☐ **Economic Impact:** Damage to infrastructure, homes, businesses, agriculture, and loss of income.
- ☐ **Environmental Impact:** Soil erosion, habitat destruction, and water pollution.

Prevention and Mitigation Measures:

- **Infrastructure:** Building dams, levees, flood barriers, and improved drainage systems.
- **Land Use Planning:** Avoiding construction in flood-prone areas and preserving natural floodplains.
- **Early Warning Systems:** Implementing advanced forecasting and early warning systems to alert communities.

- **Community Preparedness:** Educating the public on flood risks and emergency response plans.
- **Sustainable Practices:** Promoting water management practices and preserving wetlands that can absorb excess water.

Understanding the causes of floods and implementing effective management strategies can help mitigate the risks and reduce the impacts of flooding on communities and the environment.

Droughts:

Droughts are prolonged periods of deficient rainfall, causing water shortages affecting ecosystems, agriculture, water supply, and human health, with varying durations and severity.

Causes of Drought

1. **Meteorological Factors:** Lack of Precipitation, High Temperatures, Climate Patterns
2. **Hydrological Factors:** Reduced Stream flow and Groundwater Recharge, Increased Water Demand
3. **Agricultural Factors:** Poor Soil Management, Crop Choices
4. **Human Activities:** Urbanization and Industrialization, Deforestation, Water Mismanagement

Effects of Drought

- **Agricultural Impact:** Reduced crop yields, livestock losses, and increased food prices.
- **Economic Impact:** Losses in agriculture, water-dependent industries, and increased costs for water supply.
- **Environmental Impact:** Degradation of ecosystems, loss of biodiversity, increased risk of wildfires, and reduced water quality.
- **Human Impact:** Water scarcity, health problems, displacement, and reduced quality of life.

Prevention and Mitigation Measures

1. **Water Conservation**
 - **Efficient Irrigation:** Implementing drip irrigation, rainwater harvesting, and other water-saving techniques in agriculture.
 - **Domestic Water Use:** Encouraging water-saving practices such as fixing leaks, using water-efficient appliances, and reducing water wastage.
2. **Sustainable Land Management**
 - **Soil Conservation:** Adopting practices such as crop rotation, cover cropping, and no-till farming to maintain soil moisture and health.
 - **Afforestation:** Planting trees to enhance soil water retention and reduce erosion.
3. **Water Resource Management**
 - **Reservoirs and Storage:** Building and maintaining reservoirs and other water storage systems to capture and store water during wet periods.
 - **Groundwater Management:** Regulating groundwater extraction and promoting recharge techniques to maintain aquifer levels.
4. **Drought Monitoring and Early Warning Systems**

- **Data Collection:** Monitoring meteorological, hydrological, and agricultural indicators to detect early signs of drought.
 - **Forecasting:** Using climate models and weather forecasts to predict drought conditions and plan accordingly.
5. **Policy and Governance**
- **Regulations:** Implementing policies to manage water resources sustainably and mitigate the impact of drought.
 - **Education and Awareness:** Raising public awareness about drought risks and promoting community involvement in drought preparedness and response.

By understanding the causes of drought and implementing effective management strategies, communities can reduce their vulnerability to drought and enhance their resilience to future dry periods.

Earthquake:

An earthquake is a rapid, strong shaking of the ground that is frequently brought on by volcanic activity or disturbances in the Earth's crust. The magnitude and readiness of a location can determine the extent of damage and fatalities caused by earthquakes. Here are the primary causes and mechanisms of earthquakes:

Causes of Earthquakes:

1. Tectonic Plate Movements
2. Volcanic Activity
3. Human Activities like Mining, Reservoir-Induced Seismicity and Geothermal Energy Extraction

Effects of Earthquakes

- **Ground Shaking:** The primary effect, causing buildings and infrastructure to collapse.
- **Surface Rupture:** The displacement of the ground along the fault line.
- **Landslides:** Earthquakes can trigger landslides in hilly or mountainous regions.
- **Tsunamis:** Underwater earthquakes can generate tsunamis, leading to widespread coastal flooding and destruction.
- **Liquefaction:** Saturated soils can temporarily lose their strength and behave like a liquid, causing buildings to sink or tilt.

Earthquake Preparedness and Mitigation

1. **Building Codes and Standards**
 - **Seismic-Resistant Design:** Implementing construction practices that enhance the ability of buildings to withstand earthquakes.
 - **Retrofitting:** Strengthening existing structures to improve their seismic performance.
2. **Early Warning Systems**
 - **Detection and Alerts:** Using seismometers and other technologies to detect earthquakes and provide early warnings to reduce casualties and damage.
3. **Public Education and Drills**

- **Awareness Campaigns:** Educating the public on how to prepare for and respond to earthquakes.
- **Emergency Drills:** Conducting regular drills to ensure that people know what to do during an earthquake.
- 4. **Land Use Planning**
 - **Zoning Regulations:** Avoiding construction in high-risk areas such as fault lines and landslide-prone regions.
- 5. **Emergency Response Planning**
 - **Preparedness Plans:** Developing and maintaining comprehensive emergency response plans to coordinate rescue and relief efforts.

By understanding the causes and mechanisms of earthquakes and implementing effective preparedness and mitigation strategies, communities can reduce their vulnerability to earthquakes and enhance their resilience to seismic events.

Cyclones

Cyclones, also known as hurricanes or typhoons depending on their location, are intense circular storms that originate over warm ocean waters. They are characterized by low atmospheric pressure, high winds, and heavy rain. Here are key points about cyclones:

Formation:

1. **Warm Ocean Waters:** Cyclones form over ocean waters that are at least 26.5°C (80°F). The heat from the water provides the energy needed for the storm.
2. **Low Pressure System:** A low-pressure area allows air to rise and cool, leading to condensation and cloud formation.
3. **Coriolis Effect:** The rotation of the Earth causes the storm to spin. This effect is crucial for the cyclone's rotation and is more pronounced near the equator.

Types:

1. **Tropical Cyclones:** These form over warm tropical or subtropical waters.
2. **Extratropical Cyclones:** These form in higher latitudes, usually along weather fronts, and can occur over land or water.
3. **Subtropical Cyclones:** These have characteristics of both tropical and extratropical cyclones.

Impacts:

1. **Wind Damage:** High winds can cause extensive damage to buildings, power lines, and trees.
2. **Storm Surge:** The rise in sea level caused by the storm's winds pushing water onshore, leading to flooding.
3. **Heavy Rainfall:** Can lead to inland flooding, landslides, and significant damage to infrastructure.

Preparedness:

1. **Evacuation Plans:** Having a plan in place for evacuation in case of a cyclone is critical for safety.
2. **Emergency Kits:** Stocking up on essentials like water, food, medicines, and other supplies.
3. **Weather Alerts:** Staying informed through weather alerts and warnings from meteorological agencies.

Recent Notable Cyclones:

1. **Hurricane Katrina (2005):** One of the most devastating hurricanes in U.S. history.
2. **Typhoon Haiyan (2013):** One of the strongest tropical cyclones ever recorded, causing massive destruction in the Philippines.
3. **Cyclone Idai (2019):** Brought catastrophic flooding and damage to Mozambique, Zimbabwe, and Malawi.

Understanding cyclones, their formation, impacts, and safety measures can significantly help mitigate their effects on human life and property.

Landslides

Landslides are natural disasters that involve the movement of rock, earth, or debris down a slope. They can be triggered by a variety of natural and human-induced factors. Here are the main aspects of landslides:

Causes of Landslides:

1. **Natural Factors:**
 - **Heavy Rainfall:** Intense or prolonged rainfall can saturate the soil, reducing its cohesion and triggering landslides.
 - **Earthquakes:** Seismic activity can shake the ground, dislodging rocks and soil.
 - **Volcanic Activity:** Volcanic eruptions can cause landslides due to the rapid accumulation of volcanic material and the shaking of the ground.
 - **Weathering:** The gradual breakdown of rocks through weathering processes can weaken slopes over time.
 - **Snowmelt:** Rapid melting of snow can add water to the soil, increasing its weight and reducing its stability.
2. **Human-Induced Factors:**
 - **Deforestation:** Removal of vegetation reduces root strength that stabilizes the soil, making slopes more prone to sliding.
 - **Construction and Mining:** Activities that alter the landscape, such as building roads or mining, can destabilize slopes.
 - **Water Management:** Poorly managed irrigation or drainage systems can lead to water saturation of the soil.
 - **Urbanization:** Expanding urban areas into hilly regions can increase the risk of landslides due to increased load and altered drainage patterns.

Types of Landslides:

1. **Falls:** Sudden, free-fall movement of detached rocks or soil from a steep slope or cliff.
2. **Slides:** Movement of soil or rock along a distinct zone of weakness, which can be rotational (slump) or translational (planar).
3. **Flows:** Fluid-like movement of saturated soil, rock, or debris, which includes mudflows, debris flows, and earth flows.
4. **Topples:** Forward rotation of rock or soil around a pivot point below the center of gravity, leading to falling or sliding.
5. **Creeps:** Slow, gradual downslope movement of soil or rock, often imperceptible except through long-term observation.

Impacts of Landslides:

- **Human Lives:** Landslides can cause fatalities and injuries, particularly in densely populated or poorly prepared areas.
- **Infrastructure:** Damage to homes, roads, bridges, and utilities can be extensive and costly.
- **Environment:** Landslides can alter landscapes, block rivers, and destroy habitats.
- **Economy:** The economic impact includes the cost of repairs, loss of property, and disruption of local economies.

Mitigation and Prevention:

1. **Land Use Planning:** Avoiding construction on steep slopes and areas prone to landslides.
2. **Vegetation Management:** Planting trees and other vegetation to stabilize the soil.
3. **Engineering Solutions:** Building retaining walls, terraces, and drainage systems to manage water flow and stabilize slopes.
4. **Early Warning Systems:** Implementing monitoring and alert systems to provide warnings before a landslide occurs.
5. **Education and Awareness:** Informing communities about the risks and safety measures related to landslides.

Famous Landslides in India:

1.Malin landslide, Maharashtra: The **landslide** occurred on July 30, **2014**, in a village in **Malin**. The **landslide** occurred due to heavy rainfall

2.Uttarakhand Landslide: According to the Ministry of Home Affairs' Situation Report, the most recent landslides in Himachal Pradesh occurred on August 14–15, 2023, in the districts of Baddi Area, Mandi, and Shimla. Since July 9, 2023, Himachal Pradesh and Uttarakhand have been experiencing heavy rains that have resulted in flooding and landslides.

3.Irshalwadi Landslide: On July 19, 2023, a single, devastating landslide completely destroyed Irshalwadi village in the Raigad district of Maharashtra, India.